One Sum-Product Message-Passing Iteration for the Six-Bit, Two-Check Example

Abstract

This document gives every scalar used in one complete log-likelihood-ratio (LLR) message-passing iteration for the six-bit example in `p2`. It starts from the six displayed channel LLRs, forms all six variable-to-check messages, evaluates the two check nodes for every recipient, and combines the six check-to-variable messages into a-posteriori LLRs. The numerical results are computed from the rounded LLRs printed in the source slide; the extra decimal places below are computational precision, not extra measurement precision.

1 Graph, parity checks, and sign convention

Bits are indexed from zero. The parity-check matrix and the two parity-check equations are

$$H = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}, \qquad \begin{aligned} c_0 &: b_1 \oplus b_2 \oplus b_5 = 0, \\ c_1 &: b_0 \oplus b_3 \oplus b_4 = 0. \end{aligned} \tag{1}$$

Thus $\mathcal{N}(c_0) = \{1, 2, 5\}$ and $\mathcal{N}(c_1) = \{0, 3, 4\}$. Every variable node has degree one, so this cycle-free Tanner graph has two disconnected components, each containing one check node.

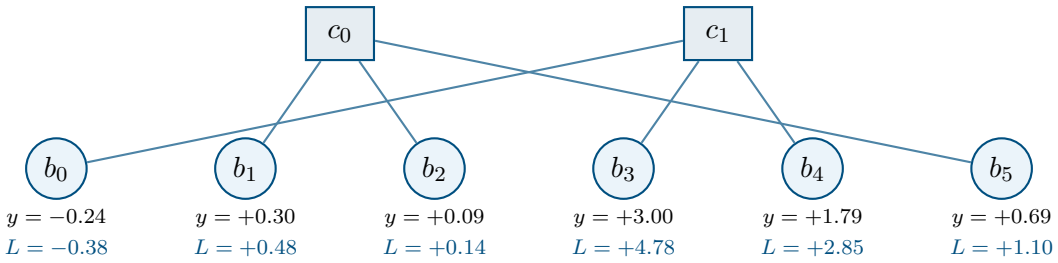


Figure 1: The six variable nodes, the two check nodes, and the displayed observations and channel LLRs.

The source example defines

$$L_i = \log \frac{P(b_i = 1 \mid y_i)}{P(b_i = 0 \mid y_i)}, \tag{2}$$

so a positive LLR favours bit one. With the bipolar mapping $x_i = 2b_i - 1$, bit one maps to $+1$. Each row of (1) requires its three bits to have even parity and therefore satisfies $\prod_{i \in \mathcal{N}(c)} x_i = -1$. This target sign is why the check update in Section 4 has a leading minus sign.

The six observations y_i and six channel LLRs L_i are both treated as given inputs. The retained example does not specify the channel parameters needed to derive the printed LLRs from the printed observations.

2 Definition of one iteration

One message-passing iteration using the flooding schedule is defined here as the following three steps:

$$U_{c\to i}^{(0)} = 0, \tag{3}$$

$$Q_{i\to c}^{(1)} = L_i + \sum_{a\in\mathcal{N}(i)\setminus c} U_{a\to i}^{(0)}, \tag{4}$$

$$U_{c\to i}^{(1)} = -2\,\mathrm{atanh}\left(\prod_{j\in\mathcal{N}(c)\setminus i} \tanh\frac{Q_{j\to c}^{(1)}}{2}\right), \tag{5}$$

$$B_i^{(1)} = L_i + \sum_{c\in\mathcal{N}(i)} U_{c\to i}^{(1)}. \tag{6}$$

Here $Q_{i\to c}$ is a variable-to-check message, $U_{c\to i}$ is a check-to-variable message, and $B_i^{(1)}$ is the a-posteriori LLR after the iteration. Because every variable has degree one, the sum in (4) is empty and

$$Q_{i\to c}^{(1)} = L_i \quad \text{on every edge.} \tag{7}$$

3 Inputs and variable-to-check messages

Table 1 contains the inputs, the initial hard decisions, and every scalar entering the check-node calculations. The initial hard decision is $\widehat{b}_i^{(0)} = \mathbb{1}[L_i \geq 0]$.

Table 1: Channel inputs and variable-to-check messages.

i	y_i	L_i	$\widehat{b}_i^{(0)}$	edge	$Q_{i\to c}^{(1)}$	$Q/2$	$t_{i\to c} = \tanh(Q/2)$
0	−0.24	−0.38	0	$0 \to c_1$	−0.38	−0.190	−0.187 746 206
1	+0.30	+0.48	1	$1 \to c_0$	+0.48	+0.240	+0.235 495 750
2	+0.09	+0.14	1	$2 \to c_0$	+0.14	+0.070	+0.069 885 890
3	+3.00	+4.78	1	$3 \to c_1$	+4.78	+2.390	+0.983 347 814
4	+1.79	+2.85	1	$4 \to c_1$	+2.85	+1.425	+0.890 637 366
5	+0.69	+1.10	1	$5 \to c_0$	+1.10	+0.550	+0.500 520 211

The transmitted word displayed in the source example is

$$\boldsymbol{b} = (0, 1, 0, 1, 1, 1). \tag{8}$$

Before message passing,

$$\widehat{\boldsymbol{b}}^{(0)} = (0, 1, 1, 1, 1, 1), \qquad H\widehat{\boldsymbol{b}}^{(0)} \bmod 2 = (1, 0). \tag{9}$$

Only bit 2 is initially wrong, and only check c_0 is violated.

4 Check-to-variable messages

For compactness, define the product used to form the message from check c to bit i:

$$p_{c\to i}^{(1)} = \prod_{j\in\mathcal{N}(c)\setminus i} t_{j\to c}^{(1)}. \tag{10}$$

Then $U_{c\to i}^{(1)} = -2\,\mathrm{atanh}(p_{c\to i}^{(1)})$. Each row of Table 2 shows both transformed inputs, their product, the inverse hyperbolic tangent, and the outgoing message. The recipient's own message is excluded from its row.

Table 2: Complete check-node arithmetic for every check-to-variable message.

edge	other bits	t_j	t_k	$p^{(1)}_{c\to i}$	$\operatorname{atanh}(p^{(1)}_{c\to i})$	$U^{(1)}_{c\to i}$
$c_0 \to 1$	$2, 5$	$+0.069\,885\,890$	$+0.500\,520\,211$	$+0.034\,979\,301$	$+0.034\,993\,577$	$-0.069\,987\,155$
$c_0 \to 2$	$1, 5$	$+0.235\,495\,750$	$+0.500\,520\,211$	$+0.117\,870\,382$	$+0.118\,420\,853$	$-0.236\,841\,706$
$c_0 \to 5$	$1, 2$	$+0.235\,495\,750$	$+0.069\,885\,890$	$+0.016\,457\,830$	$+0.016\,459\,316$	$-0.032\,918\,633$
$c_1 \to 0$	$3, 4$	$+0.983\,347\,814$	$+0.890\,637\,366$	$+0.875\,806\,306$	$+1.357\,475\,742$	$-2.714\,951\,483$
$c_1 \to 3$	$0, 4$	$-0.187\,746\,206$	$+0.890\,637\,366$	$-0.167\,213\,786$	$-0.168\,798\,923$	$+0.337\,597\,845$
$c_1 \to 4$	$0, 3$	$-0.187\,746\,206$	$+0.983\,347\,814$	$-0.184\,619\,821$	$-0.186\,761\,347$	$+0.373\,522\,694$

For example, the reply from c_0 to bit 2 excludes bit 2's own message:

$$U^{(1)}_{c_0\to 2} = -2\,\operatorname{atanh}\left[\tanh\left(\frac{Q^{(1)}_{1\to c_0}}{2}\right)\tanh\left(\frac{Q^{(1)}_{5\to c_0}}{2}\right)\right] \tag{11}$$

$$= -2\,\operatorname{atanh}[\tanh(0.48/2)\tanh(1.10/2)] \tag{12}$$

$$= -2\,\operatorname{atanh}(0.117870382293) \tag{13}$$

$$= -0.236841705808. \tag{14}$$

5 A-posteriori LLRs and decisions

Because each variable node has degree one, each bit receives one check-to-variable message, so $B^{(1)}_i = L_i + U^{(1)}_{c\to i}$. Under the convention in (2), the posterior probability, the posterior mean of x_i, and the hard decision are

$$P^{(1)}_i = P(b_i = 1 \mid \boldsymbol{y}) = \frac{1}{1 + e^{-B^{(1)}_i}}, \qquad \xi^{(1)}_i = \tanh\frac{B^{(1)}_i}{2}, \qquad \widehat{b}^{(1)}_i = \mathbb{1}[B^{(1)}_i \geq 0]. \tag{15}$$

Table 3: Combination of the channel LLR and incoming check-to-variable message.

i	L_i	incoming $U^{(1)}_{c\to i}$	$B^{(1)}_i$	$P^{(1)}_i$	$\xi^{(1)}_i$	$\widehat{b}^{(1)}_i$
0	-0.38	$-2.714\,951\,483$	$-3.094\,951\,483$	$0.043\,315\,982$	$-0.913\,368\,035$	0
1	$+0.48$	$-0.069\,987\,155$	$+0.410\,012\,845$	$0.601\,090\,959$	$+0.202\,181\,918$	1
2	$+0.14$	$-0.236\,841\,706$	$-0.096\,841\,706$	$0.475\,808\,477$	$-0.048\,383\,046$	0
3	$+4.78$	$+0.337\,597\,845$	$+5.117\,597\,845$	$0.994\,045\,276$	$+0.988\,090\,551$	1
4	$+2.85$	$+0.373\,522\,694$	$+3.223\,522\,694$	$0.961\,709\,945$	$+0.923\,419\,890$	1
5	$+1.10$	$-0.032\,918\,633$	$+1.067\,081\,367$	$0.744\,041\,476$	$+0.488\,082\,953$	1

The first message-passing iteration changes only bit 2:

$$\widehat{\boldsymbol{b}}^{(0)} = (0, 1, 1, 1, 1, 1) \quad \longrightarrow \quad \widehat{\boldsymbol{b}}^{(1)} = (0, 1, 0, 1, 1, 1). \tag{16}$$

The resulting syndrome is

$$H\widehat{\boldsymbol{b}}^{(1)} \bmod 2 = (0, 0), \tag{17}$$

so both parity checks are satisfied and the displayed transmitted word is recovered.

Because the Tanner graph is cycle-free and each variable node has only one check neighbour, this iteration already gives the exact bit marginals for the six rounded channel LLRs. Repeating the update does not change any message: each variable-to-check message remains equal to its channel LLR.

Numerical provenance. The inputs and equations come from `os00_exact_example.tex`; the retained checker is `exact_six_bit_example.py`. Every number in Tables 1–3 was independently recomputed from the rounded channel LLRs. Only sum-product quantities are included.